Sentinel Stage 2 Implementation Report

Yegor Sokolov | EECS 3311 Section A | Fall 2026

Implementation snapshot 29 October 2026 | Submission due 10 November 2026

1 Project overview and implementation status

Sentinel is a single-user Java incident-triage application for students and junior analysts investigating synthetic local security events. Shared Swing and CLI workflows connect planning, allow-listed retrieval, provenance, deterministic severity and model-assisted interpretation. Sentinel does not scan networks or execute containment actions.

All twelve Stage 1 features have connected local application paths. This report describes the working checkout, including completion work awaiting my review and publication. The latest real Qwen run produced a complete cited assessment, score 7 HIGH, a follow-up answer and an exported Markdown report and saved/reopened case. Basic verification and its limits are detailed in Section 6. Video links remain explicit placeholders; this is not yet a complete published submission.

The accepted baseline is Sentinel_Stage_1_Design_Report_V3.docx: F01-F12, UC01-UC08 and five patterns. This report follows the seven sections in Project-Stage2-Instruction.pdf; the full Project Instructions were also read. Setup instructions follow the demonstration index.

Architecture and scope

SentinelGUI and SentinelCLI submit Action values through ActionDispatcher and AgentController. AgentEngine coordinates bounded planning, retrieval and analysis. Investigation guards lifecycle, evidence and audit invariants. PlanningStrategy, ToolCommand, LLMGateway, SeverityPolicy and CaseRepository separate contracts from infrastructure. ApplicationSession injects one shared dataset, controller and executor.

The primary model remains local Qwen2.5 7B Instruct through Ollama. The optional OpenAI-compatible adapter uses the same LLMGateway boundary. Model drafts cannot choose arbitrary tools, alter fixed severity or execute response actions.

Outstanding submission work

New completion work needs my review and explicit publication approval. Demonstration videos have not been supplied. The weekly commit requirement is not yet satisfied for every period. Section 6 counts only actual published commits; future dates and unpublished files are not counted. Full KUMA behavioral validation remains Stage 3 work.

2 Feature to implementation mapping

Every feature retains its original ID and user need. Status is implemented locally with refinements. No feature or original design pattern has been silently removed. The paths below identify the principal implementation files; collaboration explanations identify each responsibility.

F01 Natural language planning

Hybrid | Original Stage 1 F01 | UC01 / SD01 | Implemented locally

AgentController creates/clarifies the case; AgentEngine delegates to PlanningStrategy; LLMPlanningStrategy and RulePlanningStrategy generate plans; PlanValidator checks tools, scopes, dependencies and limits.

Methods: start, plan, createPlan, validate, acceptRulePlan. Patterns: MVC, Strategy, Adapter.

The analyst supplies a goal, dataset, target and UTC interval. Invalid model drafts receive one repair. Missing scope leaves DRAFT, and basic rule fallback requires explicit approval on the same case. Planning never executes tools.

Files: sentinel/application/AgentController.java; sentinel/application/AgentEngine.java; sentinel/domain/Plan.java; sentinel/planning/LLMPlanningStrategy.java; sentinel/planning/PlanningStrategy.java; sentinel/planning/PlanValidator.java; sentinel/planning/RulePlanningStrategy.java.

Demonstrate in New case or Generate model plan; CLI new / plan / rule. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F01 | URL [insert accessible link] | Timestamp [mm:ss]

F02 Validated dataset import

Deterministic | Original Stage 1 F02 | UC02 / SD02 | Implemented locally

AgentController preflights state; LogImporter validates a bounded UTF-8 event array; DatasetStore publishes durable immutable versions; Investigation records version, audit and staleness.

Methods: importLogs, importFile, ingest, recordImport. Patterns: MVC.

Conflicting event IDs reject atomically; identical duplicates count once. New versions retain historical evidence, clear the old plan, invalidate sessions and stale analysis. Invalid input and failed durable writes preserve the previous dataset.

Files: sentinel/application/AgentController.java; sentinel/data/DatasetStore.java; sentinel/data/LogImporter.java; sentinel/domain/Investigation.java.

Demonstrate in Import JSON; CLI import. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F02 | URL [insert accessible link] | Timestamp [mm:ss]

Feature mapping continued

F03 Filtered security log search

Deterministic | Original Stage 1 F03 | UC03 / SD03 | Implemented locally

ToolExecutor checks a case-bound session and dispatches LogSearchCommand to DatasetStore; ResultVerifier checks canonical source payloads before Investigation attaches evidence.

Methods: openSession, execute, query, search, addEvidence. Patterns: MVC, Command.

The view submits string-valued arguments pinned to a dataset/version, target and interval. Results are ordered and deduplicated with provenance and audit references. Empty, capped and failed retrieval are distinct outcomes.

Files: sentinel/data/DatasetStore.java; sentinel/data/ResultVerifier.java; sentinel/domain/Investigation.java; sentinel/domain/Result.java; sentinel/tools/LogSearchCommand.java; sentinel/tools/ToolExecutor.java.

Demonstrate in logSearch query; CLI query --tool logSearch. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F03 | URL [insert accessible link] | Timestamp [mm:ss]

F04 Threat indicator enrichment

Deterministic | Original Stage 1 F04 | UC03 / SD04 | Implemented locally

IndicatorFeedLoader validates source-dated records; IndicatorLookupCommand delegates to DatasetStore.lookup; AgentEngine accepts only the scoped target or indicators resolved from accepted events.

Methods: loadIndicatorFeed, lookup, query, acceptEvidenceIndicators. Patterns: MVC, Command.

Normalized IP/domain lookup preserves selected feed version, verdict, source and observation date. UNKNOWN is not SAFE; stale context remains visible. Observed indicators can extend the visible plan without resetting budgets or duplicate history.

Files: sentinel/application/AgentEngine.java; sentinel/data/DatasetStore.java; sentinel/data/IndicatorFeedLoader.java; sentinel/domain/Indicator.java; sentinel/tools/IndicatorLookupCommand.java.

Demonstrate in indicatorLookup query; CLI query --tool indicatorLookup. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F04 | URL [insert accessible link] | Timestamp [mm:ss]

Feature mapping continued

F05 Account activity investigation

Deterministic | Original Stage 1 F05 | UC03 / SD03 | Implemented locally

UserActivityCommand reuses LocalQueryCommand; DatasetStore computes ordered events/counts; ActivitySummary records full interval and returned coverage; Investigation attaches unique results.

Methods: activity, execute, query, addEvidence. Patterns: MVC, Command.

The shared retrieval path reports successes, failures, hosts and unknown values. Truncation retains total-versus-returned coverage; unknown accounts return empty results. Overlapping activity/search evidence attaches once.

Files: sentinel/data/DatasetStore.java; sentinel/domain/ActivitySummary.java; sentinel/domain/Investigation.java; sentinel/domain/Summary.java; sentinel/tools/LocalQueryCommand.java; sentinel/tools/UserActivityCommand.java.

Demonstrate in userActivity query; CLI query --tool userActivity. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F05 | URL [insert accessible link] | Timestamp [mm:ss]

F06 Timeline correlation

Deterministic | Original Stage 1 F06 | UC04 / SD05 | Implemented locally

CaseEvidence resolves canonical in-scope source events; TimelineBuilder orders/deduplicates and produces relationships; AnalysisService includes the timeline in Assessment.

Methods: events, build, correlate, analyze. Patterns: MVC.

UTC order is timestamp then event ID. Shared account/host within 15 minutes creates candidate related pairs, not causation. Missing optional fields do not block ordering; source-ID conflicts reject the standalone builder. Output is capped at 1000 pairs.

Files: sentinel/analysis/AnalysisService.java; sentinel/analysis/TimelineBuilder.java; sentinel/data/CaseEvidence.java; sentinel/domain/Assessment.java; sentinel/domain/Evidence.java.

Demonstrate in Timeline after Run; CLI run / exported report. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F06 | URL [insert accessible link] | Timestamp [mm:ss]

Feature mapping continued

F07 Grounded hypothesis analysis

Hybrid | Original Stage 1 F07 | UC04 / SD05 | Implemented locally

AnalysisService prompts through LLMGateway, resolves prompt-local refs to canonical UUIDs and validates structured synthesis; Investigation stores accepted Assessment values.

Methods: analyze, validateCitations, setAssessment. Patterns: MVC, Adapter.

The model receives at most 24 canonical evidence items, fixed severity and prior hypotheses. Unknown citations, invalid schema or confidence receive one repair. Persistent failure keeps deterministic results as PARTIAL with AI analysis unavailable. Citation validity is not semantic proof.

Files: sentinel/analysis/AnalysisService.java; sentinel/domain/Assessment.java; sentinel/domain/Investigation.java; sentinel/llm/LLMGateway.java.

Demonstrate in Hypotheses after Run; CLI run. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F07 | URL [insert accessible link] | Timestamp [mm:ss]

F08 Explainable severity

Hybrid | Original Stage 1 F08 | UC04 / SD05 | Implemented locally

RuleSeverityPolicy derives Facts from source events and accepted feed context; SeverityPolicy fixes scoring; AnalysisService preserves that result independently of model assertions.

Methods: derive, assess, analyze. Patterns: MVC, Strategy.

Each rule scores once: five failed logins in ten minutes +2; subsequent success within ten minutes +3; malicious login source +2; affected critical host +1. LOW 0-2, MEDIUM 3-5, HIGH 6-8; no usable authentication means UNDETERMINED. Unknown optional facts score zero and are disclosed.

Files: sentinel/analysis/AnalysisService.java; sentinel/analysis/RuleSeverityPolicy.java; sentinel/analysis/SeverityPolicy.java; sentinel/domain/Facts.java; sentinel/domain/Severity.java.

Demonstrate in Assessment after Run; CLI run. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F08 | URL [insert accessible link] | Timestamp [mm:ss]

Feature mapping continued

F09 Response recommendations

Hybrid | Original Stage 1 F09 | UC04 / SD05 | Implemented locally

AnalysisService validates Recommendation fields, references and categories; Assessment stores accepted rationale, priority and prerequisites; ReportService formats without inference.

Methods: analyze, validateCitations, export. Patterns: MVC, Adapter.

Only COLLECT, REVIEW and ESCALATE proposals are accepted. Every proposal cites evidence and is marked for analyst approval; no containment command exists. Invalid or unavailable synthesis produces no fabricated recommendation.

Files: sentinel/analysis/AnalysisService.java; sentinel/domain/Assessment.java; sentinel/reporting/ReportService.java.

Demonstrate in Assessment recommendations; CLI run. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F09 | URL [insert accessible link] | Timestamp [mm:ss]

F10 Contextual follow up

Hybrid | Original Stage 1 F10 | UC05 / SD06 | Implemented locally

AgentEngine optionally refreshes scoped tools through RulePlanningStrategy and ToolExecutor; AnalysisService answers; Investigation owns ConversationEntry history.

Methods: ask, answer, createPlan, query, recordConversation. Patterns: MVC, Strategy, Command, Adapter.

Questions use accepted evidence, fixed severity and prior hypotheses. A refresh: prefix requests bounded collection; scope widening requires an explicit revision. Answers resolve to existing UUIDs, with one repair for invalid citations. Failures are recorded and autosaved.

Files: sentinel/analysis/AnalysisService.java; sentinel/application/AgentEngine.java; sentinel/domain/ConversationEntry.java; sentinel/domain/Investigation.java; sentinel/domain/Plan.java; sentinel/planning/PlanningStrategy.java; sentinel/planning/RulePlanningStrategy.java; sentinel/tools/ToolExecutor.java.

Demonstrate in Ask and Conversation; CLI ask. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F10 | URL [insert accessible link] | Timestamp [mm:ss]

Feature mapping continued

F11 Persistent cases and history

Deterministic | Original Stage 1 F11 | UC06-UC07 / SD07 | Implemented locally

JsonCaseRepository validates closed JSON records and case revisions; AtomicFiles replaces files atomically; DatasetStore restores retained sources/feed provenance; Investigation restores aggregate state.

Methods: save, load, list, restore, snapshot, publish. Patterns: MVC, Observer.

Locks reject competing/stale case writes. Invalid schemas/references reject before restoration. RUNNING restores as PAUSED and needs revalidation. Unavailable source versions remain inspectable but block retrieval. Case/dataset files are separately atomic, not a cross-file transaction.

Files: sentinel/data/DatasetStore.java; sentinel/domain/Investigation.java; sentinel/persistence/AtomicFiles.java; sentinel/persistence/CaseRepository.java; sentinel/persistence/JsonCaseRepository.java.

Demonstrate in Save and Open case; CLI save / open / list. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F11 | URL [insert accessible link] | Timestamp [mm:ss]

F12 Report export

Deterministic | Original Stage 1 F12 | UC08 / SD08 | Implemented locally

ReportService formats immutable accepted case data; AtomicFiles protects output; AgentController exposes preview/export to both views.

Methods: preview, export, snapshot. Patterns: MVC.

Markdown includes scope, plan, provenance/payloads, timeline/relationships, severity, hypotheses, recommendations, conversation, failures, completeness and versions. Export makes no model calls. Existing destinations require GUI confirmation or explicit CLI --overwrite.

Files: sentinel/application/AgentController.java; sentinel/persistence/AtomicFiles.java; sentinel/reporting/ReportService.java.

Demonstrate in Report and Export Markdown; CLI preview / export. Include the relevant invalid-input, partial or alternative behavior.

VIDEO PLACEHOLDER F12 | URL [insert accessible link] | Timestamp [mm:ss]

3 Implemented design patterns

MVC

SentinelGUI and SentinelCLI are views; AgentController is controller; Investigation, Plan and Assessment are model values. Both views submit the same Action to ActionDispatcher, then use shared start/import/query/run/ask/save/export services. SwingWorker separates background work from EDT rendering. This preserves Stage 1 and avoids duplicate interface logic.

Strategy

AgentEngine is context and PlanningStrategy the interchangeable port. LLMPlanningStrategy and RulePlanningStrategy implement createPlan; explicit approval selects labeled rule fallback. AnalysisService also calls SeverityPolicy.assess on deterministic Facts. Model text cannot change policy output. This matches Stage 1.

Command

ToolExecutor is invoker; ToolCommand is contract; LogSearchCommand, UserActivityCommand and IndicatorLookupCommand extend LocalQueryCommand; DatasetStore is receiver. Session.execute validates schema, scope, keys and budgets before dispatch. AgentEngine.query then verifies canonical results before attachment. This matches Stage 1.

Observer

EventBus is publisher, ProgressListener the observer contract, and both views are listeners. publish delivers immutable ProgressEvent values and isolates listener failure. CLI writes progress to stderr; Swing schedules display on the EDT. Durable AuditEntry records remain separate. This matches Stage 1.

Adapter

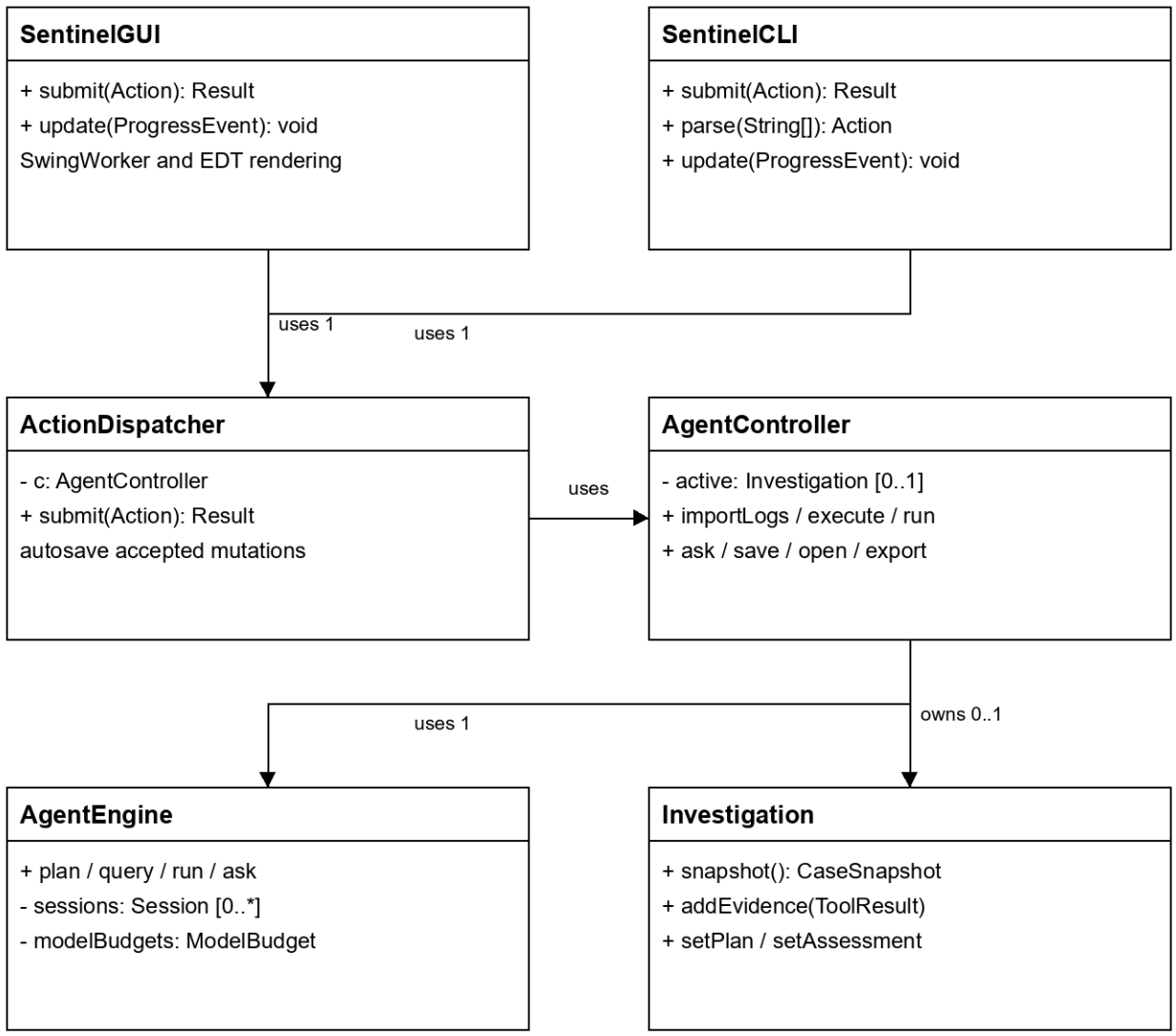
LLMGateway is target; OllamaAdapter adapts local Ollama; OpenAICompatibleAdapter optionally adapts another compatible service. complete translates typed Prompt into bounded HTTP and ModelException failures. ModelGatewayFactory selects the adapter at composition. The optional second adapter refines the planned replaceable provider boundary.

Source packages: application/, presentation/, planning/, tools/, events/, llm/ and analysis/ under src/main/java/sentinel. The following diagram views summarize the implementation. docs/uml/implemented-system.puml is the detailed editable UML class model.

Implemented interaction class view

Current implementation relationships and key methods. The full PlantUML source includes the additional command, policy, persistence and observer inheritance relationships.

Implemented interaction and application classes



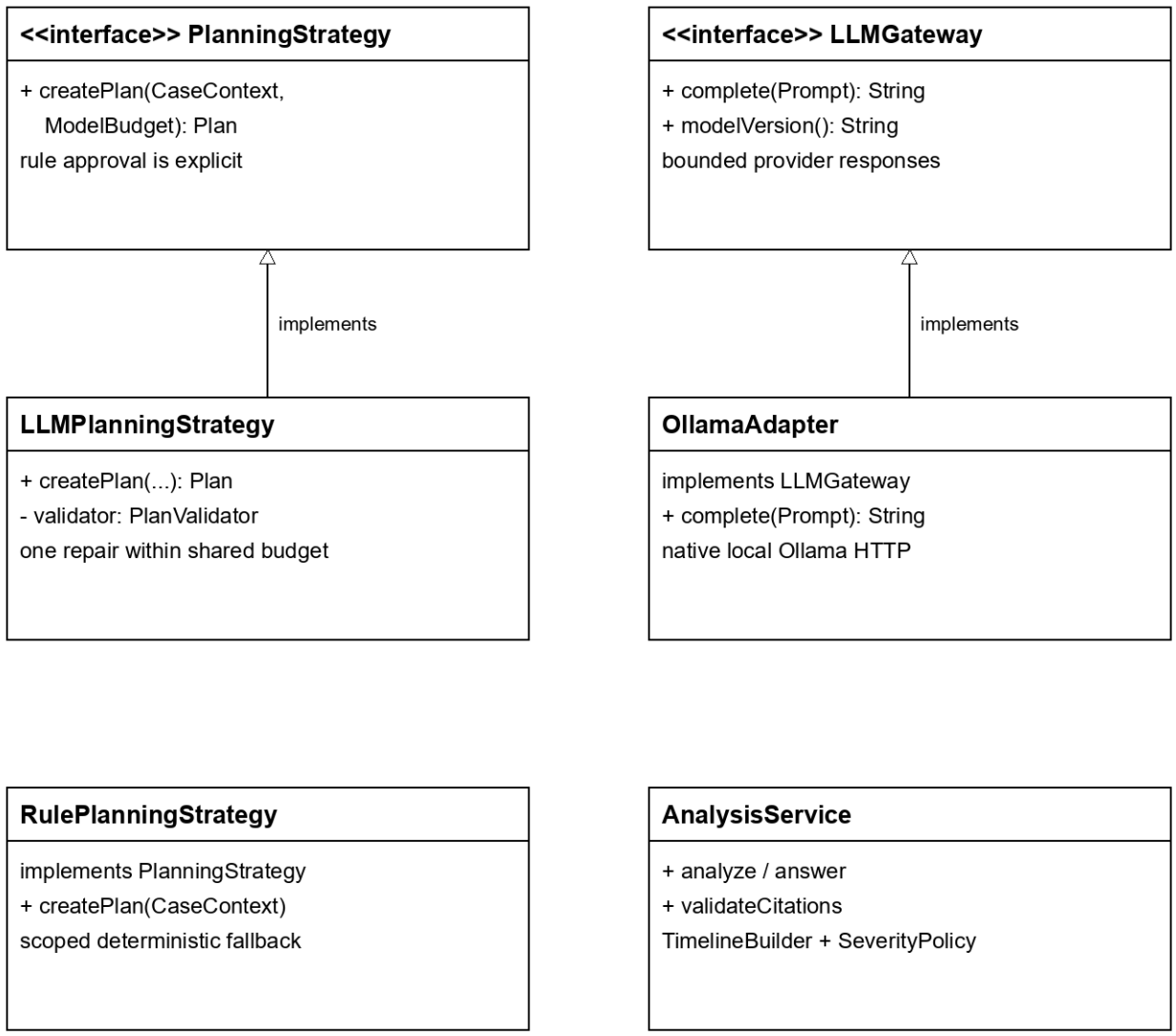
Arrow = dependency/association; hollow triangle = implements. Each service is injected once.

Current UML view interaction

Implemented planning and model class view

Current implementation relationships and key methods. The full PlantUML source includes the additional command, policy, persistence and observer inheritance relationships.

Implemented planning and model boundaries



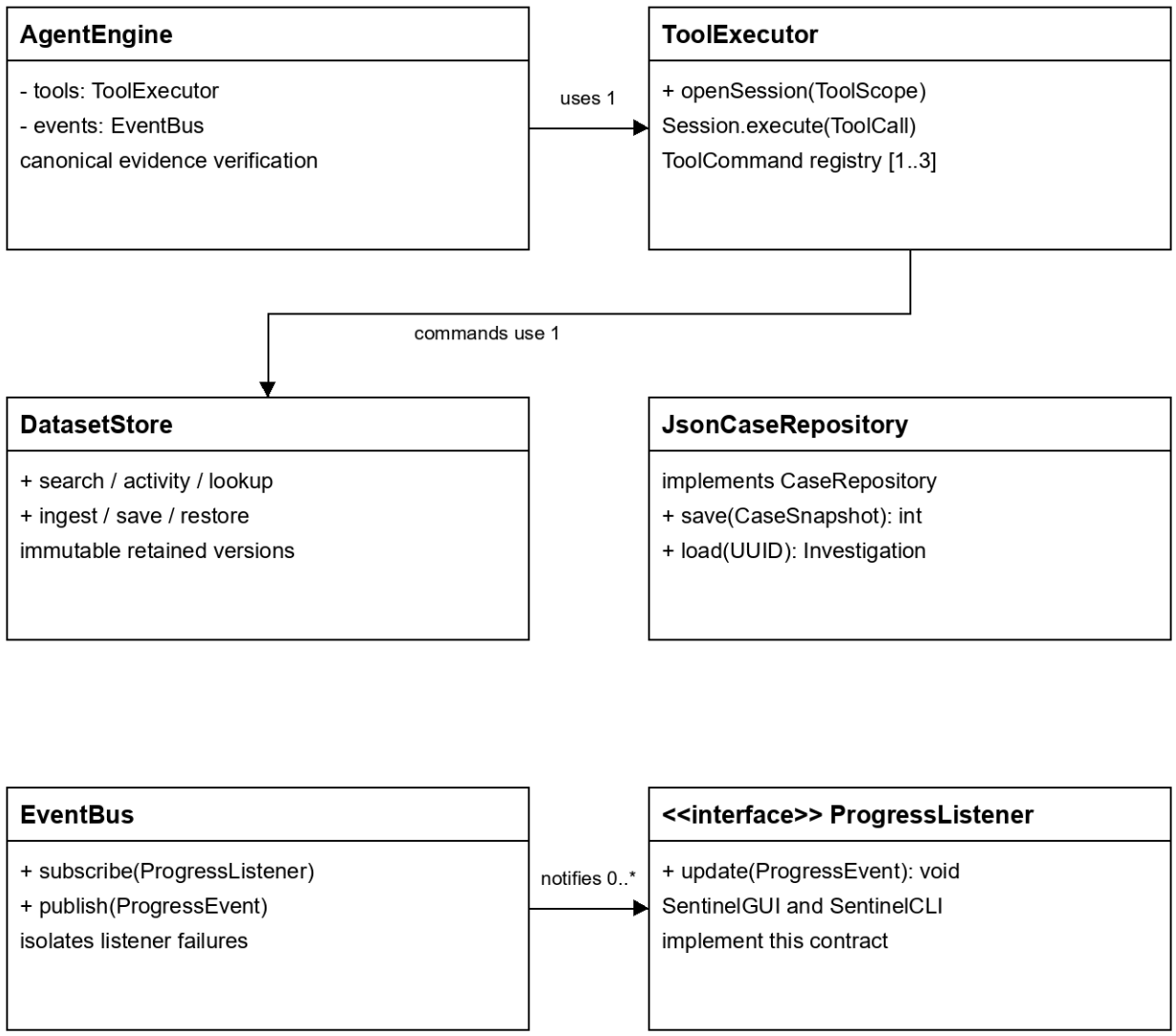
Arrow = dependency/association; hollow triangle = implements. Each service is injected once.

Current UML view analysis

Implemented data and progress class view

Current implementation relationships and key methods. The full PlantUML source includes the additional command, policy, persistence and observer inheritance relationships.

Implemented retrieval persistence and progress classes



Arrow = dependency/association; hollow triangle = implements. Each service is injected once.

Current UML view data

4 Changes and rationale

Shared controller

Original design to implemented change: ActionDispatcher and ApplicationSession route and compose both views. Reason: Remove duplicate interface business routing. Impact: UC01-UC08; application/presentation packages; MVC unchanged.

Abstract evidence boundary

Original design to implemented change: Explicit target/feed metadata, canonical ResultVerifier and case-bound sessions. Reason: Prevent forged or historical evidence from authorizing enrichment; preserve counters/history. Impact: F02-F05; CaseEvidence, CaseSnapshot, AgentEngine and ToolExecutor; Command unchanged.

Plan act observe loop

Original design to implemented change: Accepted event indicators may extend the inspectable plan within two revisions. Reason: Use observations without restarting collection. Impact: UC04/SD04; AgentEngine.run, Plan, PlanValidator; eight tool attempts and six model calls.

Grounded model synthesis

Original design to implemented change: Compact short reference labels map back to full evidence UUIDs; one bounded repair and fixed severity. Reason: Reduce output latency without weakening source validation. Impact: F07/F09/F10; AnalysisService and LLMGateway. Citations do not establish semantic truth.

Ollama infrastructure

Original design to implemented change: Optional OpenAI-compatible connector and Windows/Linux setup scripts. Reason: Permit provider changes through the existing port; keep local Qwen primary. Impact: ModelSettings, ModelGatewayFactory, adapters and scripts; Adapter preserved, no replacement feature claimed.

JSON storage

Original design to implemented change: Closed record codec, bounded UTF-8, atomic replacement, case locks and retained dataset/feed versions. Reason: Restore provenance and reject malformed/stale case state. Impact: F11; JsonCaseRepository, AtomicFiles, JsonCodec and DatasetStore. Files are separately atomic.

Run and export alternatives

Original design to implemented change: Cooperative stop, explicit partial outcomes and confirmed overwrite. Reason: Preserve accepted evidence and avoid silent replacement. Impact: UC04/UC08; AgentEngine, SentinelGUI and ReportService. Auxiliary controls refine existing features.

No Stage 1 feature or selected pattern was removed. UC01-UC08 remain the analyst goals. docs/uml/runtime-sequences.puml updates orchestration and persistence interactions; scope and failure refinements leave the use-case actors unchanged.

5 AI tools and collaboration log

I directed the incremental implementation and confirmed review/testing of Steps 1-10. My review of the later completion work remains pending. Codex generated and integrated code, prompts, tests and documentation. Its automated verification is distinguished from my own review.

Tool and model	Purpose and major tasks	Known usage and cost
Codex desktop GPT-6 family	Architecture-guided implementation, setup, domain/data services, planners, tools, adapters, analysis, persistence, orchestration, CLI/Swing, tests and report drafting. Exact model variant for every earlier session is not established.	Three US$20 top-ups = US$60. I reported 1,200 credits above the included Codex allowance, plus additional included allowance of unknown amount. Token totals and remaining balance are unknown.
Ollama 0.40.13 Qwen2.5 7B Instruct Q4_K_M	Local planning, grounded synthesis and follow-up through LLMGateway. Installed tag qwen2.5:7b. Existing GTX 1650 4 GiB; CUDA 12 backend.	No hosted API fee for local inference. Hardware, electricity and subscription allocation are unmeasured. A prior direct diagnostic measured 37 prompt / 10 output tokens; total project usage is unknown.

Credits are not tokens. US$60 records reported purchases, not a measured token-consumption cost or full project cost. No paid hosted inference or fabricated zero-token total is claimed. Further details: docs/ai-tools-and-costs.md.

Representative interactions

Evidence integration

AI contribution: Codex connected the Command invoker, canonical source validation and atomic aggregate attachment, with domain/application tests. Human contribution: I reviewed/tested Step 9 and authorized publication. Outcome: Commit 707d502; isolated publication snapshot passed 245 tests.

Rule and model planning

AI contribution: Codex implemented scope/dependency checks, explicit fallback, bounded requests and fake-server tests. Human contribution: I reviewed/tested Step 10 and authorized publication separately; later model work is not inferred to be reviewed. Outcome: Rule/model planning plus local Qwen smoke evidence, recorded separately from publication.

Completion and failure handling

AI contribution: Codex implemented timelines, rules, cited synthesis, persistence, orchestration and interfaces; it diagnosed clipping, cancellation locking and model latency. Human contribution: I requested application completion; review of new work remains pending. Outcome: Connected local workflows and automated checks, with observed real-model failures and successful retest in Section 6.

6 Development and verification summary

Verification layer	Recorded evidence
Java checks	404 tests passed with zero failures/errors/skips at the completion checkpoint, Maven Wrapper verify and JDK 17. New workflow checks cover scoring, provenance, citations, repair, shared budgets, cancellation, persistence and routing.
Earlier setup and model work	371 Java tests and 11 Python setup checks previously passed. Real native Ollama and compatible local completion/planning passed. Native Linux GPU execution and paid hosted inference were not performed.
Packaged CLI	Separate JVM invocations created/imported a synthetic case, planned, retrieved/enriched, scored, answered, exported and reopened it. Latest real observations follow below.
Actual Swing window	Saved synthetic cases, plans, partial assessment and progress were visually inspected. Clipped buttons were corrected with a grid layout. Service work runs in SwingWorker; progress is scheduled on the EDT.
Stage 3	Full coverage expansion and KUMA behavioral evaluation remain separate Stage 3 work. These Stage 2 checks are not broad agent-reliability evidence.

Latest real Qwen run: 30.54 seconds; COMPLETE; severity 7 HIGH; 1 accepted hypothesis. Timings include JVM startup and are observations, not guarantees.

Latest real follow-up: exit 0 after 26.14 seconds. Accepted cited answer.

Earlier real analysis/answer requests exceeded the 60-second deadline. Sentinel retained timeline and fixed severity as PARTIAL. Compact evidence and a shorter synthesis contract then produced a complete real-model result without raising the deadline. Retries and repairs remain budgeted; latency depends on hardware and input.

Important defects and limits

Verification corrected a feed fixture clock before its observation dates, a Swing Action import conflict, clipped controls and cancellation that read a locked snapshot. Failed follow-ups now retain an autosaved conversation failure. These findings are not an exhaustive defect claim.

Use one dataset writer per process/session. Case locks reject competing saves, but case and dataset files are not one transaction. Model context is capped at 24 accepted items, marking larger contexts partial; relationship output is capped at 1000 pairs. Critical hosts are explicitly supplied and must be resupplied in a later CLI process. Citation membership/scope is checked; semantic interpretation remains analyst-reviewable. Natural-language scope detection is conservative, not a complete intent classifier.

Weekly published commit summary

Dates are Toronto local time. Only substantive code/test commits published to origin/main are counted. Documentation-only commits and uncommitted work are excluded. Counts reflect the inspected history after the separately approved Step 10 publication. Later completion work remains unpublished.

Period	Toronto dates	Published count
1	October 20-26	5 of 5
2	October 27-November 2	5 of 5
3	November 3-9	0 of 5

Period 1

8f19cb7 Implement startup configuration and MVP verification with Maven wrapper

f99b454 Validate core domain contracts and immutable import batches

d98fadf Implement guarded case lifecycle and immutable snapshots

3a720d4 Implement atomic validated dataset import and update development records

d476e10 Implement versioned filtered log search with source-linked evidence and tests

Period 2

8a8866e Implement versioned account activity with counts, coverage and tests

49f1976 Implement validated versioned indicator lookup with source evidence and tests

3125f77 Implement scoped tool execution with budgets, cancellation and tests

707d502 Integrate active-case imports and verified evidence attachment

cdd850f Implement approved rule planning with scoped plan validation

Period 3

No qualifying commits yet. This period is in the future relative to the report snapshot. Genuine code/test progress and separately authorized publication are still required.

Step 10 was reviewed, tested and published as cdd850f on October 29. Its isolated publication snapshot passed 272 tests and both component demos. Period 2 now satisfies its five-commit minimum. Extra October commits cannot satisfy November 3-9; that future period still requires genuine code/test progress and publication.

7 Demonstration index

VIDEO PLACEHOLDERS: supply recordings, accessible URLs and direct timestamps after recording the real application. Screenshots and test results do not replace the required feature videos. Collectively the recordings must show both GUI and CLI.

Feature	Suggested segment	Video URL and timestamp
F01 Natural language planning	New case or Generate model plan; CLI new / plan / rule	[insert accessible URL] [mm:ss]
F02 Validated dataset import	Import JSON; CLI import	[insert accessible URL] [mm:ss]
F03 Filtered security log search	logSearch query; CLI query --tool logSearch	[insert accessible URL] [mm:ss]
F04 Threat indicator enrichment	indicatorLookup query; CLI query --tool indicatorLookup	[insert accessible URL] [mm:ss]
F05 Account activity investigation	userActivity query; CLI query --tool userActivity	[insert accessible URL] [mm:ss]
F06 Timeline correlation	Timeline after Run; CLI run / exported report	[insert accessible URL] [mm:ss]
F07 Grounded hypothesis analysis	Hypotheses after Run; CLI run	[insert accessible URL] [mm:ss]
F08 Explainable severity	Assessment after Run; CLI run	[insert accessible URL] [mm:ss]
F09 Response recommendations	Assessment recommendations; CLI run	[insert accessible URL] [mm:ss]
F10 Contextual follow up	Ask and Conversation; CLI ask	[insert accessible URL] [mm:ss]
F11 Persistent cases and history	Save and Open case; CLI save / open / list	[insert accessible URL] [mm:ss]
F12 Report export	Report and Export Markdown; CLI preview / export	[insert accessible URL] [mm:ss]

Setup and short review workflow

JDK 17 and the Maven Wrapper build the application. First use downloads Maven 3.9.9 and pinned dependencies. Windows local Qwen setup uses scripts/setup-qwen.ps1; Linux uses scripts/setup-qwen.sh. Native Linux execution has not been verified on this Windows host. Keep .env local and ignored; copy .env.example only when .env does not already exist.

```
$env:JAVA_HOME = 'C:\Program Files\Java\jdk-17'
.\mvnw.cmd --batch-mode verify
java -jar target/sentinel-0.1.0-SNAPSHOT.jar --check-config
java -jar target/sentinel-0.1.0-SNAPSHOT.jar gui
```

CLI returns a JSON Result envelope with status and message; structured payloads inside message are JSON text. Progress is on stderr. Exit codes are 0 success, 2 invalid input, 3 operational failure. Supply --case UUID on later invocations. Successful mutations save automatically. A partial run is a successful operation with an explicit PARTIAL assessment. A restored PAUSED case uses resume --case UUID to revalidate its unchanged plan before Run; revision 2 remains revision 2 and no model request is spent.

```
java -jar target/sentinel-0.1.0-SNAPSHOT.jar new --dataset local --request "Investigate demo" --target demo-analyst --start 2026-10-22T12:00:00Z --end 2026-10-22T13:00:00Z --feed demo-feed-v1 --rule
java -jar target/sentinel-0.1.0-SNAPSHOT.jar import --case UUID --file data/scenarios/account-takeover.json
java -jar target/sentinel-0.1.0-SNAPSHOT.jar rule --case UUID
java -jar target/sentinel-0.1.0-SNAPSHOT.jar run --case UUID
java -jar target/sentinel-0.1.0-SNAPSHOT.jar ask --case UUID --question "Explain the accepted evidence"
java -jar target/sentinel-0.1.0-SNAPSHOT.jar export --case UUID --out runtime/exports/case.md
```

GUI review: create a case, import the takeover fixture, approve a rule plan or generate a model plan, Run, inspect every result tab, ask a follow-up, save/open, and export. The takeover fixture scores 7 HIGH or 8 when workstation-1 is explicitly declared critical. Model-offline behavior must remain partial; absent analysis never means benign.

References: README.md; docs/feature-implementation-mapping.md; docs/design-patterns.md; docs/implementation-architecture.md; docs/verification-log.md; docs/ai-collaboration-log.md; docs/ai-tools-and-costs.md; docs/uml/implemented-system.puml. Assignment sources: Project-Stage2-Instruction.pdf, Project-Instruction Full.pdf, and the accepted Stage 1 V3 report.